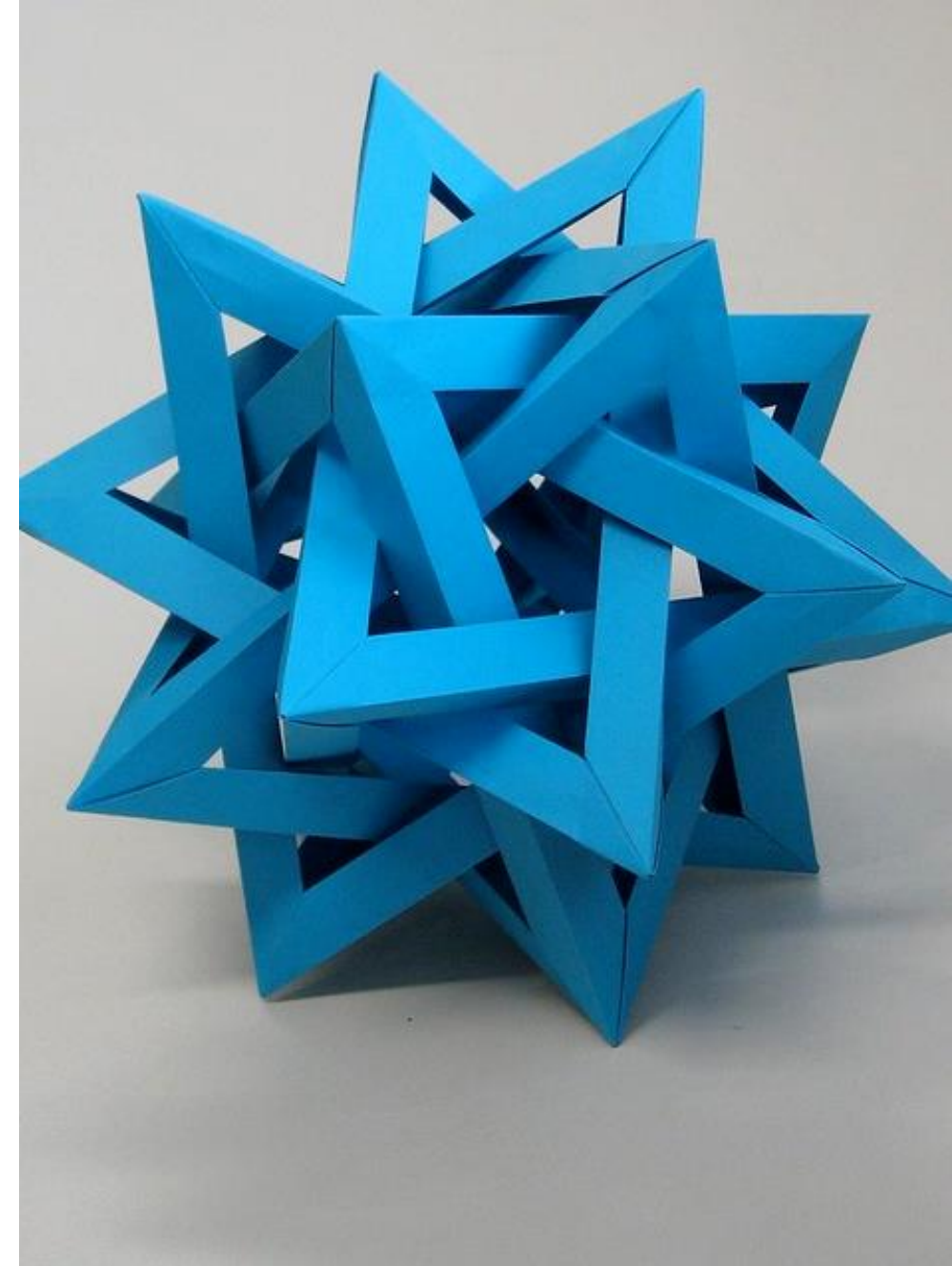




LAB 6

FUNCTIONS



General Suggestion

- Try to wrap all your programs in a **main()** function, as seen in class.

06.1.1 – Count Vowels

- Read request from Lab text..

06.1.1 – Count Vowels

- **Pattern seen in class**: count characters that match a condition

06.1.2 – Word Count

- Read request from Lab text..

06.1.2 – Word Count

- First Approach:
 1. Remove the initial and final spaces (if any). This can be done with the **.strip()** method of strings.
 2. Count the remaining spaces: another instance of «counting the characters that match a certain condition»
- **Alternative:** if you are already familiar with lists, you should know that there's a string method that returns a **list of substrings** based on a separator.
- **Try both!!**

06.2.1 – NGO

- Read request from Lab text..

06.2.1 – NGO

- The solution is a combination of stuff we've already seen:
 - if statements with **multiple branches** , with complex conditions using **boolean operators**
 - Iterated acquisition of data with a **sentinel value** to terminate
 - .. and of course, functions definition and usage.

06.2.2 – Roman Numerals

- Read request from Lab text..

06.2.2 – Roman Numerals

- Break down in the problem in sub-problems
 - Define a **convertDigit(r)** function taking a string of length 1 containing a single roman digit **r** and returning the corresponding value

06.2.2 – Roman Numerals

- Define a **romanToDecimal(r)** function taking a string representing a **multi-digit** roman number, and returning the corresponding decimal value. This function will iteratively:
 - Apply the algorithm explained in the exercise text on the **first two digits**, of the roman string, to update the decimal value. This step will call **convertDigit(r)** multiple times, to get the two digits values.
 - Eliminate 1 or 2 characters (depending on the case) **from the beginning** of the string → **How can we eliminate characters from a string? Isn't it immutable?**
 - Repeat until the string is empty.